

RF Admittance level switch Rope Type for Solid Application, Model: RFLS-300SR

PRODUCT DESCRIPTION:



The RFLS-300R RF Admittance type level switch is a simple, low cost, rugged point Level Switch designed for Level detection of Bulk materials in a bins or silo. This level switch can be used for detecting high, low level or for differential level detection and is used in various level detection applications, cement industries, material handling plants, chemical industry, power utilities industry, food industry, pulp/paper industry, pharmaceuticals industry, water/ treatment plants, foundry industry etc. The specialty of the probe is that it tends to ignore the build-up on the probe, thus making it suitable for application, which tends to stick on the probe. The unit has options with integral and Remote electronics along with time delay feature, LED indication, fail safe mode etc.

SALIENT FEATURES:

- Wide range of industrial applications.
- Easy installation and maintenance.
- Weatherproof / Flameproof enclosure as per requirement.
- PTFE-insulated electrode capable of withstanding temperatures up to 180°C.
- No moving parts – no mechanical wear and tear.
- Adjustable differential time delay with ON/OFF timing.
- Fail-Safe High / Low selection – field configurable.
- Ignores build-up of sticky and conductive material between the probe and vessel wall.
- Flexible Rope-Type Probe suitable for deep silos and hoppers.
- Top Mounting for convenient installation.
- Suitable for Powder, Granule and Bulk Solid Materials.
- Adjustable sensitivity for reliable detection of different materials.
- Integral / Remote Electronics available as per application.
- Customised Probe Length available as per vessel height.
- Heavy-Duty Construction suitable for demanding industrial applications.
- Reliable level detection even with material build-up on the probe



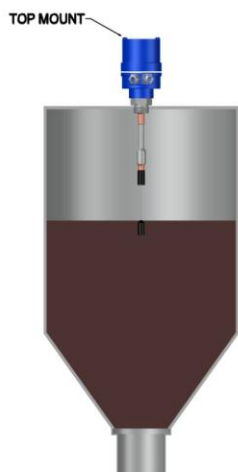
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OPERATING PRINCIPLE:

The RFLS-300R RF Admittance Level Switch operates on the principle of Radio Frequency (RF) Admittance Measurement. The probe and vessel wall form an electrical sensing system. The electronic circuit continuously applies an RF signal to the probe and monitors the electrical admittance between the probe and vessel ground. When the solid material reaches the sensing probe, the dielectric properties around the probe change, resulting in a change in RF admittance. The instrument detects this change and, when it exceeds the preset sensitivity level, activates the output relay to provide a reliable High-Level or Low-Level indication. The Rope/Flexible Probe is designed for Top Mounting and is particularly suitable for deep silos, hoppers, bins and storage vessels. The Build-Up Compensation / Coating Rejection feature minimises the effect of sticky, conductive or coating material accumulated on the probe, ensuring reliable level detection. Fail-Safe High/Low Selection, adjustable sensitivity and ON/OFF time delay allow the instrument to be configured according to different material and process requirements. With no moving parts, the instrument provides reliable long-term operation with minimum mechanical wear and maintenance.

PROBE CONSTRUCTION SUITABLE FOR MATERIAL BUILD-UP

LEVEL SWITCH INSTALLATION



The RFLS-300SR a conventional probe may tend to malfunction or give false level signals when material accumulates on the probe. The COAT Guard electrode is specially designed to overcome this problem by using advanced RF build-up sensing technology. The COAT Guard electrode is electrically insulated from the active sensing probe and is driven with the same RF frequency and polarity as the active electrode. This arrangement creates a controlled RF field around the sensing probe. When material build-up or coating forms on the probe, the COAT Guard circuitry detects and compensates for the build-up, preventing the accumulated material from being interpreted as an actual level signal.

The compensated build-up region is effectively isolated from the active sensing measurement. Therefore, material coating on the probe does not normally cause false switching. When the actual material level rises and reaches the active sensing probe, the RF admittance changes beyond the compensated build-up condition. The electronic circuit detects this change and switches the output to indicate the actual material level.

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The COAT Guard construction provides reliable operation in applications involving sticky, conductive, powdery, granular and coating-forming materials.

The Rope Type Flexible Probe is suitable for Top Mounting and is particularly useful for deep silos, hoppers, bins and storage vessels.

COMPONENTS OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE

ELECTRICAL CONNECTION UNIVERSAL /230 VAC/110VAC/24 VDC

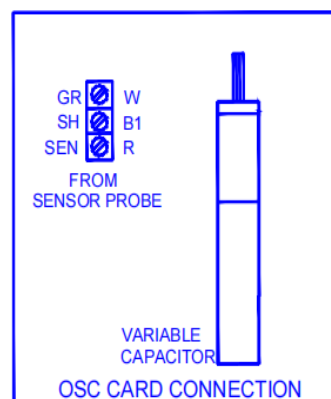
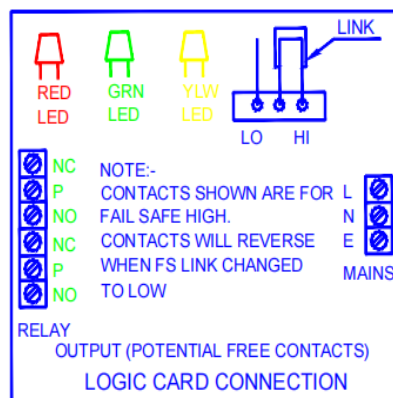
Integral Type



Remote Type



Controller /RF Cable/ Remote Probe



DATA SHEET

RFLS-300 – RF Admittance Level Switch

Selection / Ordering Information

Selection	Code	Description
Model No.	RFLS-300	RF Admittance Level Switch
Application	SR	Solids Rope Type
Mounting Type	IN/RE	Integral / Remote
Probe Type	RD / RP/HD	Rigid Rod / Rope Flexible/Heavy Duty type Probe
Insulation	FP/PT/ CE	Full PTFE / Part PTFE / Ceramic
Probe Diameter	D1 / D2	Standard 16mm / Heavy Duty 30mm
Probe Material	04 /4L/ 16/ 6L/MS/ SM	SS304/SS304L / SS316/ SS316L /Mild steel/ Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED



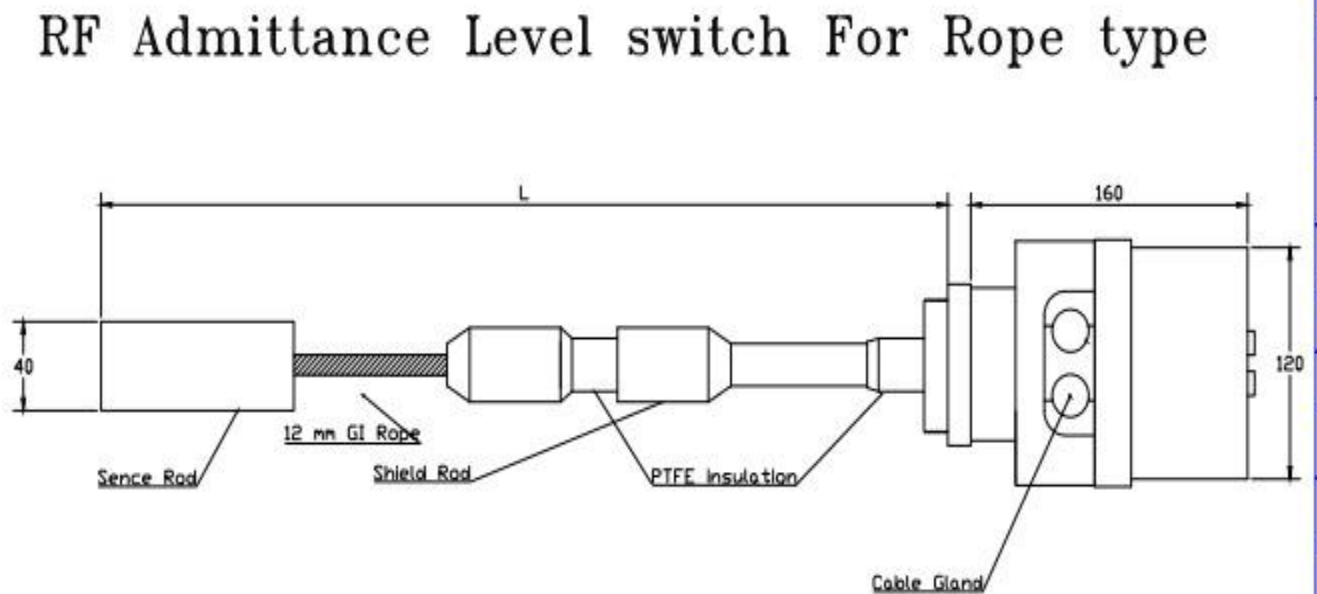
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Selection	Code	Description
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
MOC-Process Connection	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customised
Temperature	T1/ T2/TS	Max -20–150°C / 200°C/Customer Specified Special Designed
High Temperature stand off	H8/H1/H2/H3/HS	80MM/150 MM /200MM /300 MM /Special standoff Extension
Remote Cable	05 / 10 / 15 / 20 / 25 / 30/SR	5 / 10 / 15 / 20 / 25 / 30 m/ Special Customer Requirement
Cable Gland	G1/ G2 / G3/ G4	PG13.5 / ½"NPT / ¾" ET/ PG16
Ground / Reference	GS / GX	Standard / Special
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	500 mm to Customized

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Mechanical Drawing: Integral Rope Type:



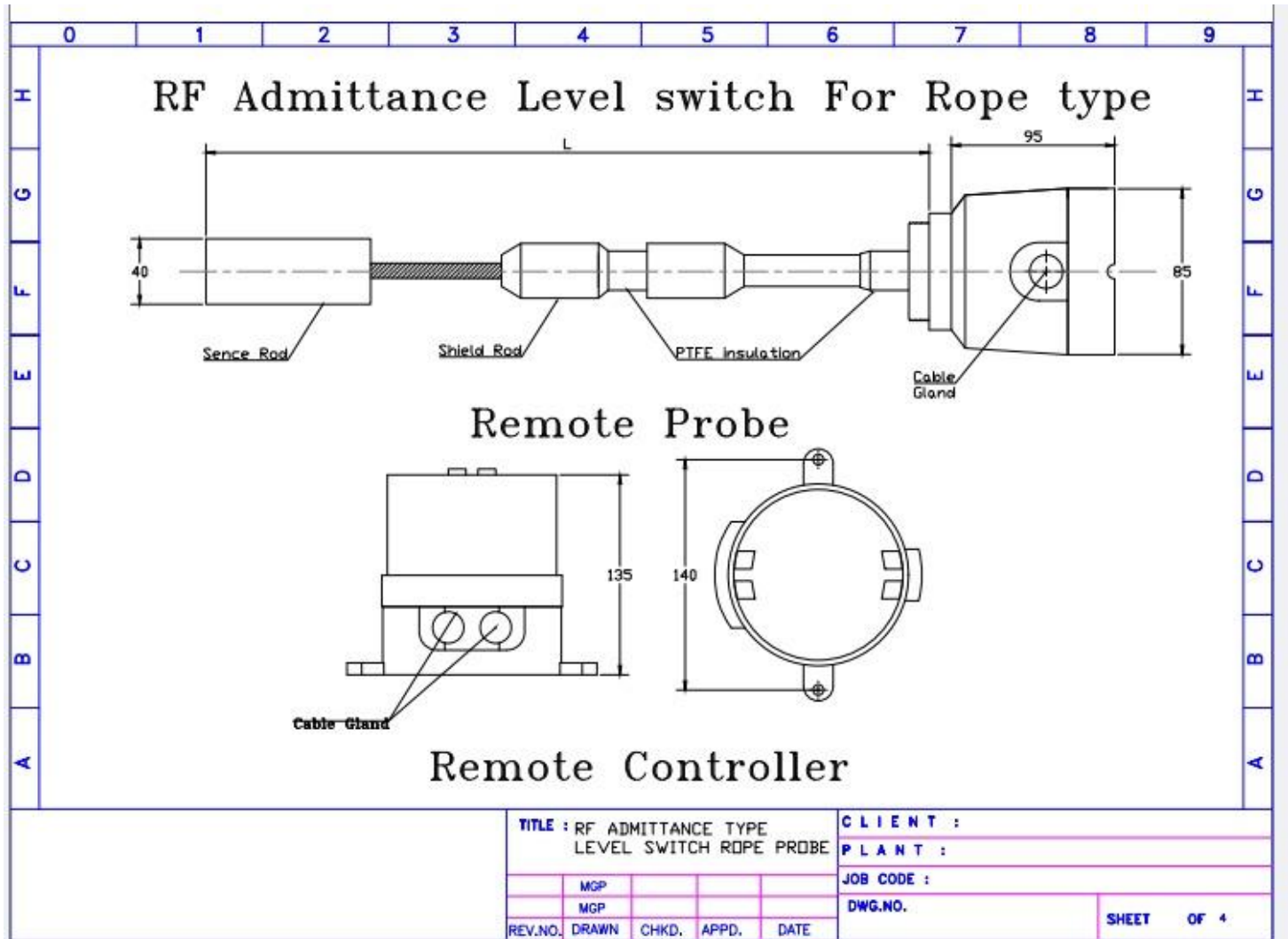
RF Admittance Level switch
Rope Type
Integral Probe
L : Insertion Length

TITLE : RF ADMITTANCE TYPE LEVEL SWITCH ROPE PROBE				
	MGP			
	MGP			
REV.NO.	DRAWN	CHKD.	APPD.	DATE

CLIENT :
PLANT :
JOB CODE :
DWG.NO.
SHEET OF 4

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Mechanical Drawing: Remote Probe Type:



Manufactured By:

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